

GIS Background Information

Here's a list and explanation of important features in ArcView:

Project: In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

View  : A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

Table of Contents: All themes in a View are listed to the left of the map. The Table of Contents shows the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

Theme: A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

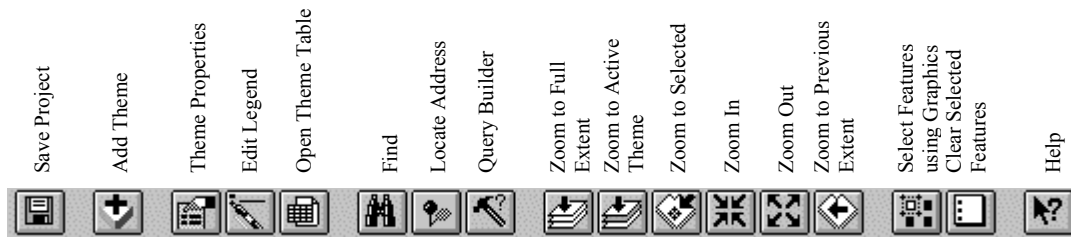
Tables  : Information formatted in rows and columns. A component of an ArcView project used for displaying tabular data.

Layout  : The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

Hotlink: In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

ArcView Buttons Reference Guide

View - Buttons



View - Tools

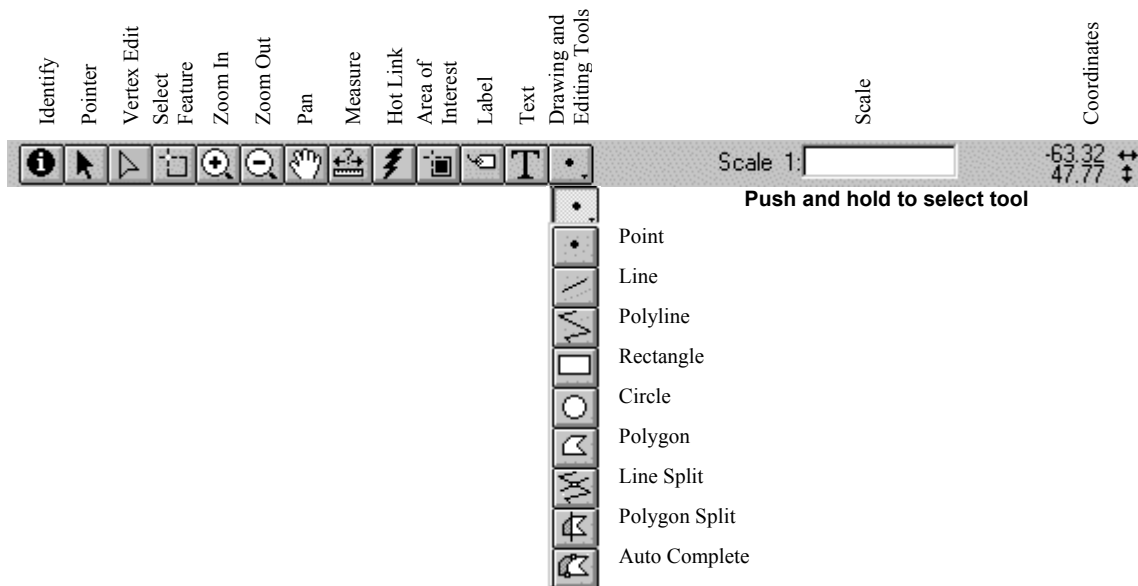
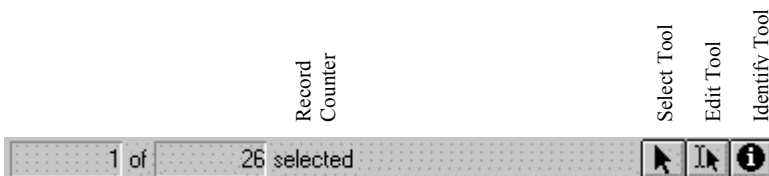


Table - Buttons



Table - Tools



2.0 Chart - Buttons

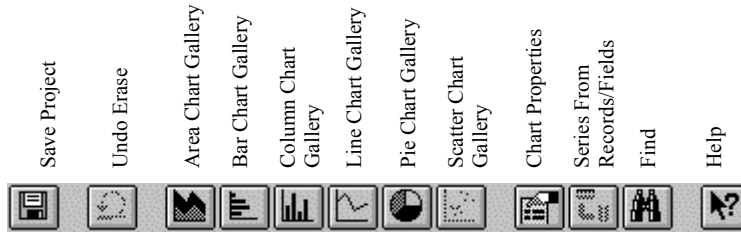
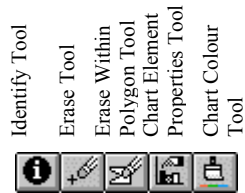
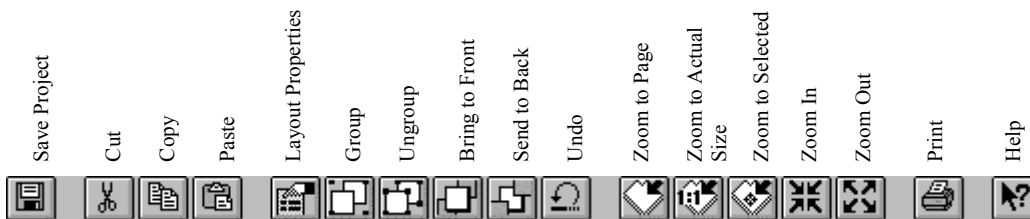


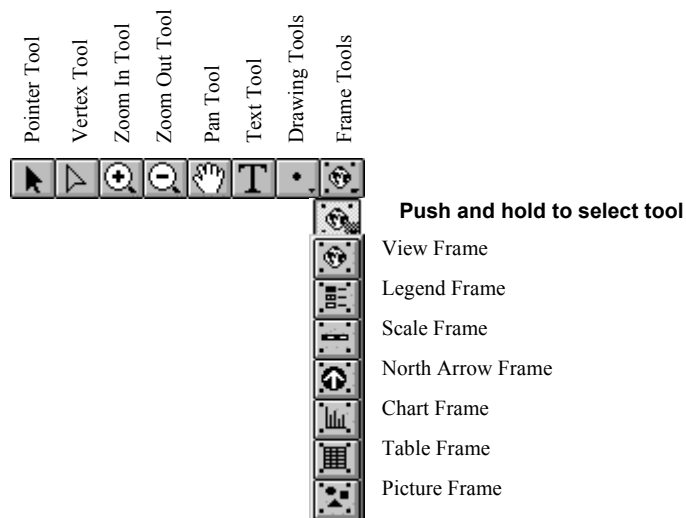
Chart - Tools



Layout - Buttons



Layout - Tools



For more information: www.esricanada.com

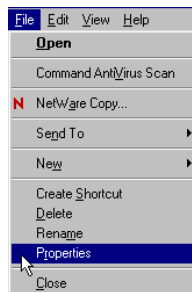
Getting Started

Before you begin, the Parks lesson data folder must be **copied** into a drive in which you the student has writing and editing capabilities. To do this use My Computer / Windows Explorer / or a file manager program. This will probably be the drive containing your own personal account on the school's system or the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete.

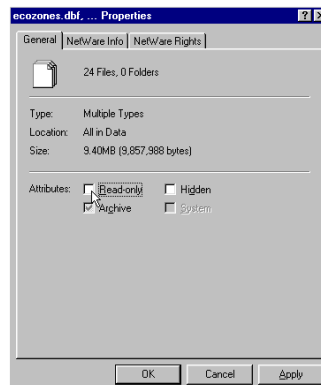
Once the Park's data folder has been **copied** to the required destination you will have to follow the following instructions closely to prepare the data for use:

1. Open your new Park's data folder
2. Choose **edit > select all** (they will all appear Blue)

3. From the menu choose:
File > Properties



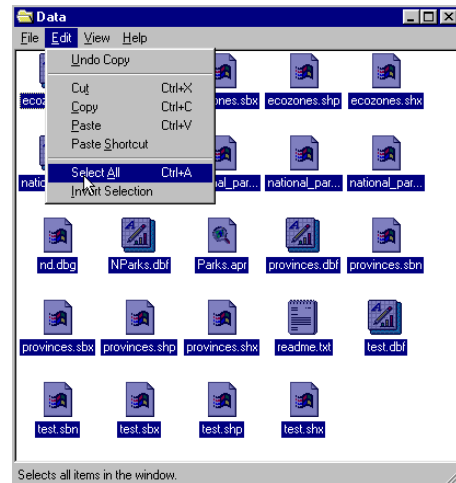
4. Remove the check mark in the box beside the label **read only**.



5. Click **apply**, then **OK**

The data in your parks data folder can now be opened edited and saved using ArcView to carry out the following project.

- !** Denotes an observation.
- ?** Denotes a question to be answered on the Student Activity Sheet.



Starting the Project

The purpose of this lesson is to learn how to use data tables to display information in a meaningful way as map and chart presentations.

STARTING Arc View

Begin by starting up the **Arc View** program.

1. Open Arc View.
2. Click *as a blank project*.
3. Click **O.K.**



The Arc View program will launch. If necessary, click the window maximize button so that the Arc View window fills the screen.

When the program starts, it automatically opens a project window titled "Untitled."

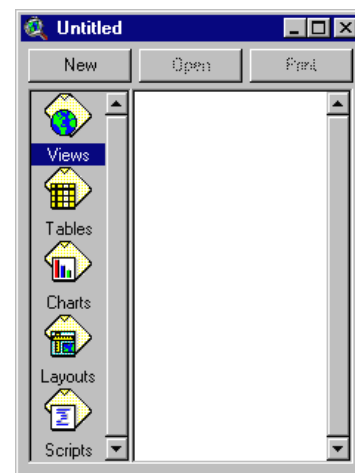
Arc View uses **projects** to group together and organize maps that are used together. A project has now been created for you to use with this activity.

4. Click the window maximize button on the project window.
5. Start a new view by clicking **New** while "Views" is highlighted. If necessary, click the window maximize button to maximize the view window.

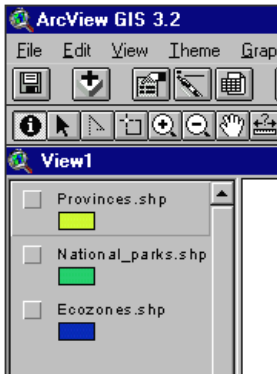
6. Click **Add theme button**  and use the shift key to select from the data folder just created:

National_Parks.shp
Ecozones.shp
Provinces.shp

7. Click **O.K.**

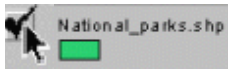


This will add these three themes to the table of contents of the new view. They are now available for display and editing.



8. Click and drag the **Province.shp** theme down to the bottom of the table of content.

9. Check mark the **Provinces** theme  to make it visible.

10. Check mark the **National Parks** theme  to make it visible.

ARCVIEW - DISPLAYING MAP LEGENDS

1. Double click **National Parks** theme.

By changing the colour ramp on a legend from **shades of a single colour** to a **full spectrum**, we can better distinguish the classes.

2. From **legend editor** box, click **legend type** list, select **graduated colour**.

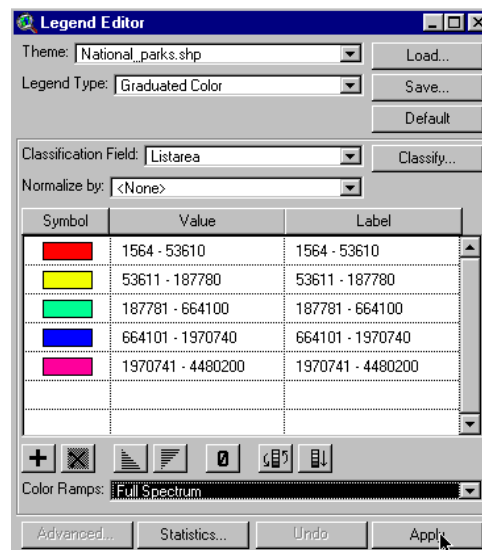
3. Click **classification field** list, select **listarea**.

4. Click **colour ramp** list, select **full spectrum**.

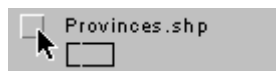
5. Click **apply** and close the **legend editor** window.

The National Parks are now grouped into classes based on area.

? Does this make it easier to see the classes?



6. Un-check **province** theme.



? Can you see the parks better?

What can you say about size distribution of the parks?

What is the reason for this pattern?

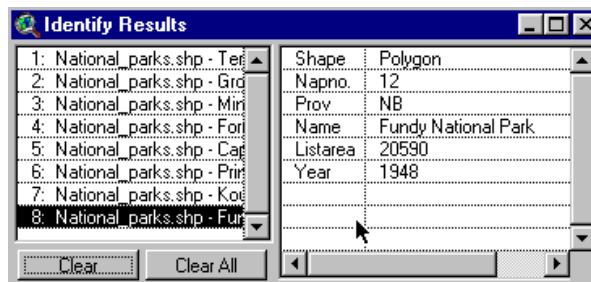
ARCVIEW - USING DATA TABLES ATTACHED TO A MAP

Using data tables, we will use the link between the tables and the map to identify national parks.

1. Check mark the **provinces** theme again.
2. Click the **zoom in**  tool.
3. Click and hold down the left mouse button and draw a box around the Atlantic Provinces
4. Click on **National Park** theme in the table of contents to make it active (the theme will now appear raised).
5. Click the **identify** tool. 
6. Place the cursor over one of the parks and click the left mouse button.

! Note the information table for the selected park.

7. Check each park in Nova Scotia, New Brunswick, Prince Edward Island and Newfoundland to build an **identify results table**. You should find 8.
8. Turn off the province theme to find the two small ones (1 PEI, 1 NS).
9. Click each park listed in the left-hand table.
11. Record the *name* and *listarea* for each park.



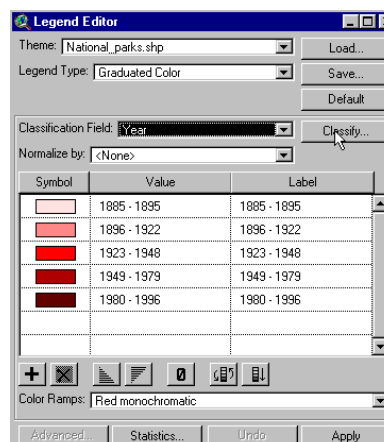
? Fill in Table 1 for the Atlantic Provinces. List the name and the area of the Parks in the table provided. Start with the largest park and list in descending order to the smallest park. Use the *listarea* to compare sizes.

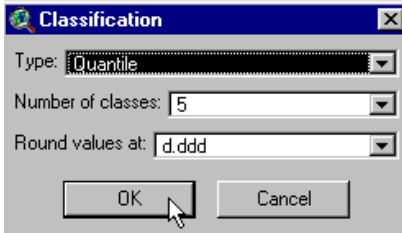
? Perform the same test on a different area of Canada. Record these results in table 2.

ARCVIEW - USING DATA FOR QUANTILE CALCULATIONS

By looking at year of establishment data, we will conclude where parks were first established and the pattern of new park designations.

1. Close the **identify results** window.
2. Click **zoom to full extent** button.  Double click **National Parks** theme.
3. Open **classification field** list, select **year**.
4. Click **classify** button.





5. Open **type** list, select **quantile**.
6. Click **O.K.**
7. Open **colour ramp** list, select **full spectrum**.
8. Choose **Apply** and close the **legend editor** window.

9. Turn off **provinces** theme

? Where are most of the new parks? Why?

10. Turn the province theme on to identify the province into which they fall.
11. Make the **National_Parks.shp** theme your active theme by clicking on it once. You should now see it raised.

12. Click **open theme/ table** button 

13. Click the title of the **year** field to select it



14. Click **sort ascending** button  to arrange the records in ascending order by year of establishment.

? Which is the oldest national park and when was it established? Which are the newest national parks and when were they established?

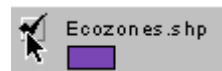
! Note that those parks that are reserves do not have establishment dates. Also note that park names are listed several times, once for each polygon within the park. (Park areas are sometimes cut by lakes or rivers or contain islands, creating several polygons to depict a single park).

15. Close **table** window.
16. Turn off the **National Parks** theme and **province** theme

ARCVIEW - QUERYING LINKED DATA ON A MAP

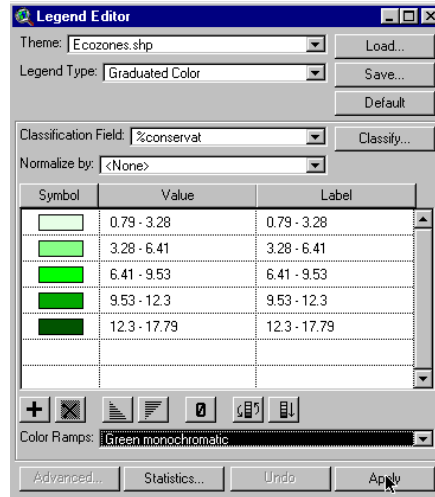
Now that we can display data through colours on a map, we will learn to query additional data linked to a map.

1. Open **Ecozone** theme alone by clicking to make it visible.
2. Double click **Ecozone** theme to edit.



3. Open **legend type** list, select **graduated colour**.
4. Open **classification field** list.
5. Select **%conservat** field (this is the percent of ecozone area covered by all conservation area types).

6. Open **colour ramp** list, select **green monochromatic**.



7. Click **apply**.

8. Close **edit legend** window.

9. Click **identify** button  and build an **identify results table** for selected ecozones. To select more than one Ecozone hold down the shift button as you select ecozones. Note that each Ecozone that you click on the left side of the table has a table of its own that shows on the right side with data about it. Look here to answer the questions below.

**? Which three ecozones have the highest percent total conservation area?
Which two have the lowest?**

ARCVIEW - COMPUTING NEW DATA

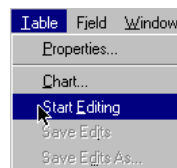
We will take data we have on national parks and total protection to calculate the amount of conservation area in each ecozone that is not a national park.

1. Make the *ecozones.shp* theme your active theme (it will appear raised).

2. Click **open theme/ table** button. 

3. From the menu select **table**.

4. Select **start editing**.



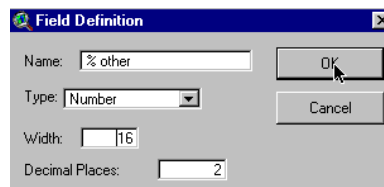
5. Click menu **edit**.

6. Select **add field**.

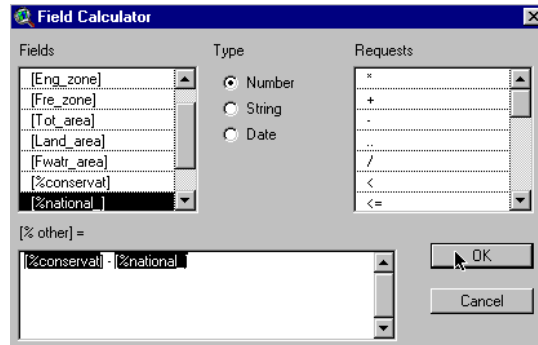
7. Change *name* to *%other*.

8. Change **decimal place** to **2**.

9. Click **O.K.**



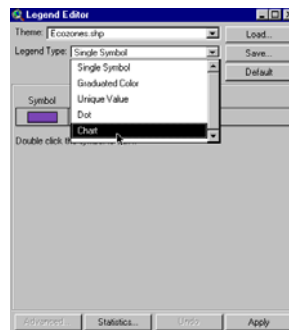
10. Click **calculate** button.
11. Build **expression** to create the mathematical equation to calculate the % conservation area minus the % national park area by selecting **field**, double click **%conservat**, then choose **requests**, double click - double click **field %National** . See the expression created.
12. Click **O.K.**



13. Click menu **table**.
14. Select **stop editing**.
15. Click **yes** to save edits.
15. Close **table** window.

ARCVIEW - DISPLAYING DATA WITH CHARTS

1. Double click **ecozone** theme.
2. Open **legend type** list.
3. Select **chart**.



4. Click **%National** field.
5. Click **add**.
6. Repeat for **%other** field.
7. Click **pie chart** type.
8. Double click **background symbol** and change colour to a light shade by clicking on **paintbrush** and selecting a new colour.



9. Click **properties** button.
10. Change **maximum size** to **16**.
11. Click **O.K.**
12. Click **apply**.
13. Close **legend edit** window and **colour palette** window.

14. Click on **identify** button. 

! Note that only one chart is applied to each ecozone, therefore ecozones with more than one polygon, such as Taiga Shield, have non-assigned areas.

? Find the name of the ecozone with all of its conservation area represented by national parks.

? Find the name of the ecozone with has very little national park representation. Based on this analysis, which ecozone should be considered for new national parks?

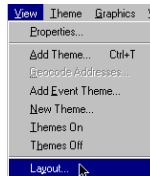
! You have completed your examination of National Parks and Conservation Areas. Now create a layout for printing from any section of this analysis or create your own analysis for printing.

ArcView - Map Layout for printing

Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

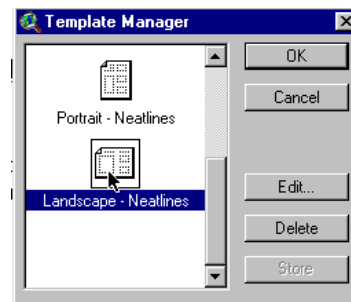
SET UP MAP

1. Click menu **view**,
2. Click **layout**



3. Select the **layout template** you want to use (one of the landscape templates is suggested)

4. Click O.K.
5. Maximize layout screen



EDIT TITLE

1. Double click on the title *View1*
2. Edit the title to reflect the purpose of the map, e.g., and population
3. Click O.K.
4. Resize the title using the corner handles and move it into a position that is visually balanced
5. Reposition legend if necessary.

forestry,

EDIT NORTH ARROW

1. Double click on the *north arrow* and select a different presentation. Ensure that that the arrow is pointing in the representative north direction of the map
2. Reposition or rotate (using rotation function) as needed

EDIT SCALE BAR

1. Double click *scale bar*
2. Choose a new style of scale bar
3. Change the units to be used
4. Change the number of scale units to be used
5. Change the number of unit divisions on the left of the scale to be used.
6. Find a combination of style, scale and divisions that is visually pleasing
7. Position to an appropriate location.

ADD LABELS, TEXT, OTHER INFORMATION

1. Click on the **text** tool
2. Click at the position on your layout where you want the text to appear
3. In the **text properties** box, enter your text, e.g., "produced by (your name, date)"
4. Select desired formatting information such as **alignment, spacing and rotation**
5. Click O.K. when it appears, as you want it
6. Click on **pointer** tool
7. Point to new text
8. Click once to change location or size, Click twice to change content

! Experiment with all of the components of the layout until you have the look you want.

CHANGE THE NAME OF THE LAYOUT

1. Click menu **layout**
2. Select **properties**
3. Change the "**name**", Click O.K.

ARRANGING YOUR LAYOUT

- !** Experiment with the layout buttons (use the tools page provided at the beginning of this lesson)
- to **group** (tie pieces together so they move as one)
 - to **ungroup** (break apart the ties so that individual pieces move)
 - **bring to front** (change order that pieces were laid down, e.g. lines on top of polygons)
 - **send to back** (change order that pieces were laid down, e.g., polygons behind lines)
 - **zoom** (make image bigger or smaller)
 - **neat line** (border around layout)

PRINT MAP LAYOUT

1. Click **print** button
2. Click **setup**
3. Select your printer properties as outlined by your teacher
4. Click O.K. to print

! **Note** that the map can also be saved to a printer file.